

EDUCATION

- **Purdue University – West Lafayette, IN, USA** **2023 – Present**
Ph.D. in Mechanical Engineering
- **University of Michigan – Ann Arbor, MI, USA** **2020 – 2022**
M.S. in Robotics
- **Purdue University – West Lafayette, IN, USA** **2014 – 2020**
B.S.E. in Mechanical Engineering

RESEARCH EXPERIENCE

- **School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA** **8/2023 – Present**
Graduate Research Assistant
- **Department of Mechanical Engineering, University of Michigan – Ann Arbor, MI, USA** **6/2022 – 5/2023**
Research Associate
- **Robotics Department, University of Michigan – Ann Arbor, MI, USA** **1/2021 – 5/2022**
Graduate Research Assistant
- **School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA** **5/2019 – 5/2020**
Undergraduate Research Assistant
- **Polytechnic Institute, Purdue University – West Lafayette, IN, USA** **5/2015 – 5/2016**
Undergraduate Research Assistant

TEACHING EXPERIENCE

- **Purdue University – West Lafayette, IN, USA** **1/2026 – Present**
Teaching Assistant: Instructed ML tutorials for graduate students (ME597) “Industrial IoT Implementation for Smart Manufacturing”
- **Purdue University – West Lafayette, IN, USA** **1/2015 – 5/2025**
Teaching Assistant: Instructed ML tutorials for graduate students (ME597) “Industrial IoT Implementation for Smart Manufacturing”

RESEARCH INTERESTS

- Robotics and Automation
- Computer vision and Artificial intelligence
- Cyber-physical system
- Smart manufacturing

MENTORING, OUTREACH, AND LEADERSHIP ACTIVITIES

- **Summer Undergraduate Research Fellowships (SURF) program**
Purdue University – West Lafayette, IN, USA
 - 01. Yoonseo Lee (B.S.E in Mechanical Engineering) **2025**
- Topic: Safety-aware high-dimensional motion planning
 - 02. Richard Li (B.S. in Computer Science) **2024**
- Topic: Optimization-based robotic grasp detection
- **Mentoring Undergraduate and Master Students**
School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA

01. Trevor Terrill Kates (B.S.E in Mechanical Engineering) - Topic: Robot-integrated cell-based manufacturing toward physical AI	2026 – Present
02. Daniel Sungwoo Bae (M.S in Mechanical Engineering) - Topic: AI-integrated autonomous sealing robot	2026 – Present
03. Junho Sohn (M.S in Mechanical Engineering) - Topic: Interpretable AI for human motion analysis in manufacturing tasks <i>(Journal submission)</i> Sohn, J., Lee, H., Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M. (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. <i>Journal of Manufacturing Systems</i> .	2025 – Present
04. Yoonseo Lee (B.S.E in Mechanical Engineering) - Topic: Safety-aware high-dimensional motion planning	2025
05. Evin Lugo Li (B.S.E in Mechanical Engineering) - Topic: Adaptive grasp planning for suction and two-finger grippers <i>(Conference proceeding)</i> Lee, H., Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. In <i>International Manufacturing Science and Engineering Conference</i> (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers.	2024 – 2025
● Multidisciplinary Design Program (MDP) University of Michigan – Ann Arbor, MI, USA	
01. Yuhang Ning (M.S in Electrical and Computer Engineering) - Topic: Autonomous environment mapping for periodically reconfigured environments	2022
02. Jiwen Jia (M.S in Electrical and Computer Engineering) - Topic: Visual servo control design for a mobile manipulator	2021

FELLOWSHIPS, HONORS, AND AWARDS

● Best Poster Award U.S. – Korea Conference (UKC) 2025, Atlanta, GA, USA	2025
● <i>Italian Machine Tool Technology Award</i> Italian Trade Agency, Italy	2025
● <i>NSF Student Travel Support Award</i> ASME International Manufacturing Science and Engineering Conference (MSEC) 2025 SME North American Manufacturing Research Conference (NAMRC) 53, Clemson University – Greenville, SC, USA	2025
● <i>Young Professional NET Award</i> The Korean Federation of Science and Technology <i>(Conference presentation)</i> Sim, Y. ^{†*} , Lee, J., Lee, H., Han, C., Lee, H., & Jun, M. (2025, August 17-20). <i>I2DTM: Immersive and Interactive Digital Twin Metaverse Framework</i> [Conference Presentation]. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States. <i>(Journal publication)</i> Jeon, J., Sim, Y., Lee, H., Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. <i>Journal of Manufacturing Systems</i> , 79, 504-514.	2024 – 2025
● <i>Best Poster Award</i> DigiTwin 2024 for Manufacturing, Sustainability, Safety, and Resilience	2024
● <i>Winkelman Fellowship</i> Purdue University – West Lafayette, IN, USA	2023 – 2024

- *DSCD and AACC Student Travel Awards* 2022
The 2022 Modeling Estimation and Control Conference (MECC), Jersey City, NJ, USA.
- *Multidisciplinary Design Program (MDP) Summer Fellowship* 2021
University of Michigan – Ann Arbor, MI, USA

JOURNAL PUBLICATIONS

- | ‡: Equal contribution | *: Corresponding author |
|---|--|
| 01. Sim, Y., Kim, E.* | Lee, H., Sohn, J., & Jun, M. (2025). Structured Sound Data Augmentation Based on Spectral Analysis for Machine Tool Operational State Monitoring. <i>The International Journal of Advanced Manufacturing Technology</i> , 141, 1705-1719. doi: https://doi.org/10.1007/s00170-025-16797-w |
| 02. Kim, E., Lee, H., Sim, Y., Lee, J., & Jun, M.* | (2025). Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction. <i>CIRP Annals – Manufacturing Technology</i> , 74, 579-583. doi: https://doi.org/10.1016/j.cirp.2025.04.082 |
| 03. Han, C., Gabor, T., Lee, H., Lee, J., Akin, S., & Jun, M.* | (2025). Pulsed Cold Spray System for Physical Unclonable Function Generation. <i>Procedia CIRP</i> , 137, 176-180. doi: https://doi.org/10.1016/j.procir.2025.02.269 |
| 04. Jeon, J., Sim, Y., Lee, H., Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* | (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. <i>Journal of Manufacturing Systems</i> , 79, 504-514. doi: https://doi.org/10.1016/j.jmsy.2025.01.01 (#1 in Operations Research & Management Science ^{OOIR 2025}) |
| 05. Lee, H.* | Toner, T., Tilbury, D., & Barton, K. (2025). GraspMixer: Hybrid of Contact Surface Sampling and Grasp Feature Mixing for Grasp Synthesis. <i>IEEE Transactions on Automation Science and Engineering</i> , 22, 10920- 10936. doi: 10.1109/TASE.2025.3530795 (#12 in Automation & Control System ^{OOIR 2025}) |
| 06. Lee, J., Akin, S.* | Sim, Y., Lee, H., Kim, E., Nam, J., Song, K., & Jun, M.* (2024). A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing. <i>Manufacturing Letters</i> , 41, 1515-1525. doi: https://doi.org/10.1016/j.mfglet.2024.09.178 |
| 07. Han, C., Lee, J., Lee, H., Sim, Y., Jeon, J., & Jun, M.* | (2024). Zero-shot Autonomous Robot Manipulation via Natural Language. <i>Manufacturing Letters</i> , 42, 16-20. doi: https://doi.org/10.1016/j.mfglet.2024.09.067 |
| 08. Lee, H.* | Toner, T., Tilbury, D., & Barton, K. (2022). Multi-sensor aided deep pose tracking. <i>IFAC-PapersOnLine</i> , 55(37), 326-332. doi: https://doi.org/10.1016/j.ifacol.2022.11.205 |

JOURNAL IN PROCESS

- | ‡: Equal contribution | *: Corresponding author |
|---|--|
| 01. (In preparation) Lee, H., Sim, Y., Lee, H., Lee, K., Jun, M., & Lee, J.* | (2026). BRIDGES: Bi-directional Robotic Integration with Digital-twin for Generalized, Efficient, and Safe Automation in Manufacturing. |
| 02. (In preparation) Sim, Y., Lee, H., Lee, H., Paek, S., Lee, J., Yoon, B., & Jun, M.* | (2026). MMRL: A Multi-Modal Avatar based Remote Interaction and Localization Framework for Digital Twin Cross-Validation in Smart Manufacturing. |
| 03. (Under review, submitted in Mar. 2026) Lee, H., Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M.* | (2025). PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Validation. <i>Robotics and Computer-Integrated Manufacturing</i> . |

04. [\(Under review, submitted in Mar. 2026\)](#) Lee, H., Sim, Y., Yoon, B., Lee, H., Park, H., Jang, J., Kim, C., & Jun, M.* (2026). Toward provably safe AI-driven robotics autonomy for manufacturing processes: A review of methodologies and future directions. *Journal of Manufacturing Systems*.
05. [\(Under review, submitted in Jan. 2026\)](#) Sohn, J., Lee, H., Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M.* (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. *Journal of Manufacturing Systems*.
06. [\(Under review, submitted in Jan. 2026\)](#) Jun, M.*, Sim, Y., Kim, E., Lee, H. (2026). Open-world condition recognition for discrete manufacturing. *CIRP Journal of Manufacturing Science and Technology*.
07. [\(Under review, submitted in Dec. 2025\)](#) Nam, J., Lee, H., Han, C., Athinarayanan, R., Jun, M., & J. Lee.* (2025). C³AM: Conversational CAD/CAM Automation via Multimodal Large Language Model Agents. *Journal of Manufacturing Systems*.
08. [\(In second round of review\)](#) Lee, H., Han, C., Gabor, T., Sim, Y., Akin, S., Jun, M., & Lee, J.* (2025). Manufacturable Physically Unclonable Identifiers via Cold Spray: A Full-Stack Framework with Implicit Neural Representation for Scalable Product Authentication. *Journal of Intelligent Manufacturing*.
09. [\(In second round of review\)](#) Lee, H., Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). GRAVITON: Generalized Resampling Algorithm for Visual Inference in Target Object Localization. *Computers in Industry*.

PEER-REVIEWED CONFERENCE PROCEEDINGS

‡: Equal contribution

*: Corresponding author

01. [\(In preparation\)](#) Lee, H., Sim, Y., Lee, H., Yoon, B., Lee, J., Garcia, J. E. N., Rastgaar, M., Jang, J., & Jun, M.* (2026, November). High-Precision 3D Trajectory Estimation Using Wire Encoder based Module with String Dynamic Model Compensation. In 2026 International Mechanical Engineering Congress and Exposition (IMECE). American Society of Mechanical Engineers.
02. [\(In preparation\)](#) Jun, M.*, Sim, Y., Lee, H., Kim, E., & Shakouri, A. (2026, October 11-16). *Acoustic Fingerprinting for Online Condition Monitoring in Progressive Stamping*. International Conference on the Technology of Plasticity 2026, Jeju Island, Republic of Korea.
03. [\(Under review, submitted in Mar. 2026\)](#) Yoon, B., Lee, H., Sim, Y., & Jun, M.* (2026, September). LiDAR-Integrated Coarse-to-Fine Optimization for Geometrically Consistent Triangle Splatting from Mobile Robots. In 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE.
04. [\(Accepted in Jan. 2026\)](#) Nam, J.*, Lee, H., Han, C., Athinarayanan, R., Jun, M., & J. Lee. (2026, June). Gen-Cad/cam: Generative and Interactive Cad/cam Framework for Simulation-Informed Adaptive Maintenance. In *International Manufacturing Science and Engineering Conference*. American Society of Mechanical Engineers.
05. [\(Accepted in Jan. 2026\)](#) Sim, Y.*, Kim, E., Sohn, J., Barr, J., Lee, H., Beach, B., Jun, M. & Shakouri, A. (2026, June). Generative Latent Space Modeling of Stamping Sound for Anomaly Detection and Predictive Maintenance. In *International Manufacturing Science and Engineering Conference*. American Society of Mechanical Engineers.
06. [\(Under review, submitted in Jan. 2026\)](#) Han, C., Sim, Y., Jung, H., Lee, J., Lee, H., Kang, Y., Woo, S., Kim, G., Park, H., & Jun, M.* (2026, July). IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer. In *Knowledge Discovery and Data Mining 2026 (KDD)*. ACM.
07. Lee, H.*, Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October). EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning. In 2025

08. Sim, Y.*, Kim, E., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June). Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A006). American Society of Mechanical Engineers. doi: <https://doi.org/10.1115/MSEC2025-155613>
09. **Lee, H.***, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers. doi: <https://doi.org/10.1115/MSEC2025-155673> (NSF Student Travel Support Award)
10. Han, C.*, **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June). A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine. In *International Manufacturing Science and Engineering Conference* (Vol. 88100, p. V001T02A003). American Society of Mechanical Engineers. doi: <https://doi.org/10.1115/MSEC2024-124383>

CONFERENCE PRESENTATIONS

- | ‡: Presenter | *: Corresponding author |
|--|-------------------------|
| 01. Jun, M.‡*, Lee, H. , Sim, Y., Lee, H., Yoon, B., & Bae, D. (2026, June 28- July 3). <i>Leveraging a Vision Foundation Model for Adaptable and Robust Object 3D Localization and Manipulation under Real Shop-Floor Environments</i> . International Conference on Precision Engineering and Sustainable Manufacturing 2026, Singapore. | |
| 02. Lee, H.‡* , Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M. (2026, January 29-30). <i>PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Estimation</i> . 2026 Institute for Control, Optimization and Network Student Research Conference, West Lafayette, IN, United States. | |
| 03. Lee, H.‡* , Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October 19-25). <i>EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning</i> . 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems, Hangzhou, China. | |
| 04. Kim, E., Lee, H. , Sim, Y., Lee, J., & Jun, M.‡* (2025, August 17-23). <i>Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction</i> . 74th CIRP Annals – Manufacturing Technology, Stockholm, Sweden. | |
| 05. Sim, Y.‡*, Lee, J., Lee, H. , Han, C., Lee, H., & Jun, M. (2025, August 17-20). <i>I2DTM: Immersive and Interactive Digital Twin Metaverse Framework</i> . International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States. | |
| 06. Lee, H. , Sim, Y., Han, C., Lee, J., & Jun, M.‡* (2025, July 6-11). <i>ARMPASS: Autonomous Robot Motion Planning with Adaptive Safe Set</i> . International Conference on Precision Engineering and Sustainable Manufacturing 2025, Chiang Mai, Thailand. | |
| 07. Sim, Y.‡*, Kim, E., Lee, J., Lee, H. , Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June 23-27). <i>Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition</i> . International Manufacturing Science and Engineering Conference 2025, Greenville, SC, USA. | |
| 08. Lee, H.‡* , Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June 23-27). <i>GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers</i> . International Manufacturing Science and Engineering Conference 2025, Greenville, SC, USA. | |

09. Han, C.[‡], Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025, June 1-4). *Pulsed Cold Spray System for Physical Unclonable Function Generation*. CIRP International Symposium on Electro Physical and Chemical Machining 2025, Vancouver, BC, Canada.
11. Jun, M.^{**}, Lee, J., Kim, E., Sim, Y., Han, C., & **Lee, H.** (2024, July 7-12). *Generalized and Generative AI for Smart Manufacturing*. International Conference on Precision Engineering and Sustainable Manufacturing 2024, Busan, Republic of Korea.
12. Han, C.^{‡*}, **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June 17-21). *A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine*. International Manufacturing Science and Engineering Conference 2024. Knoxville, TN, USA.
13. **Lee, H.**^{‡*}, Toner, T., Tilbury, D., & Barton, K. (2022, October 2-5). *Multi-sensor aided deep pose tracking*. Modeling, Estimation and Control Conference 2022, Jersey City, NJ, United States.

POSTER SESSIONS

01. Han, C., Sim, Y., Lee, J., **Lee, H.**, & Jun, M. (2025, August 5-9). Industrial Audio Foundation for Agile Machine Listening. U.S. – Korea Conference 2025, Atlanta, GA, United States. **(Best Poster Award)**
02. Sim, Y., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E. (2024, December 2). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition*. Manufacturing And Materials Research Laboratories Symposium 2024, West Lafayette, IN, United States.
03. Lee, J., Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., & Jun, M. (2024, October 15-16). *Generative AI Based Real-Time Human-Data Interaction for Smart Manufacturing*. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States. **(Best Poster Award)**
04. **Lee, H.**, Sim, Y., Lee, J., Han, C., & Jun, M. (2024, October 15-16). *Digital Twin & Vision Guided Autonomous Robotic Path Planning*. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States.
05. Lee, J., Kim, E., Han, C., **Lee, H.**, & Sim, Y. (2024, March 24-26). *Generalized and Generative Smart Manufacturing*. North Academy of Engineering Regional Meeting 2024, West Lafayette, IN, United States.
06. Han, C., **Lee, H.**, & Jun, M. (2024, May 24-26). *Robotics in Smart Manufacturing*. Congressional Artificial Intelligence Caucus 2024, Washington, D.C., United States.

PATENTS

01. Jun, M., Lee, J., **Lee, H.**, Akin, S., Han, C., Gabor, T., & Sim, Y. (2025, September 18). *Cold spray-enabled physically unclonable identifier and its spectral authentication via implicit neural representation* (U.S. Patent Application No. [Number Assigned]). U.S. Patent and Trademark Office.

PROFESSIONAL SOCIETY MEMBERSHIPS & SERVICES

- | | |
|---|----------------|
| • Reviewer of the IEEE Transactions on Automation Science and Engineering | 2025 – Present |
| • Reviewer of the IEEE/RSJ International Conference on Intelligent Robots and Systems | 2025 – Present |
| • Reviewer of the ASME Manufacturing Science and Engineering Conference | 2025 – Present |
| • Member of the Institute of Electrical and Electronics Engineers (IEEE) | 2025 – Present |
| • Member of the American Society of Mechanical Engineers (ASME) | 2025 – Present |
| • Member of the Society of Manufacturing Engineers (SME) | 2025 – Present |
| • Graduate Association for Manufacturing Excellence / Treasurer | 2025 – Present |
| • Korean-American Scientists and Engineers Association (KSEA) / Member | 2024 – Present |

INDUSTRIAL PROJECTS

- **Genesem, Inc.** 2026 – Present
 - Development of a CAD-based part localization in a semiconductor equipment for part replacement.
- **Hitachi America, Ltd.** 2024 – Present
 - Developed a multi-modal feature extraction method to fuse data from a glove sensor and a camera.
 - Extended the feature extraction module to perform motion analysis and task segmentation of human operators during manufacturing processes.
- **BCI Solution, Inc** 2024 – 2025
 - Implemented a digital twin of FANUC CRX-10iA/L using MuJoCo physics simulator and Python.
 - Established bi-directional communication among FANUC CRX-10iA, its digital twin, and CESMII platform for automation and verification of core-making operations.
- **Korea Institute of Machinery and Materials (KIMM)** 2024 – Present
 - Developed a generalized sampling algorithm to extend a cutting edge object segmentation AI to 6-DOF object pose estimation method for machine tending.
 - Developed a hybrid computer vision pipeline to localize metal debris inside CNC machine that must be removed.
- **BCI Solution, Inc** 2023 – Present
 - Built a server-client system using Python and KAREL to communicate FANUC controller via external PC.
 - Integrated 6-DOF object pose estimation algorithm with FANUC controller via the server-client system for tool-path adaptation
- **Terra Drive Systems (TDS)** 2023 – 2023
 - Installed a stethoscope-based sound sensor and a mini PC to establish a data pipeline.
 - Developed a sound recognition AI model to monitor status of Makino a81 CNC machine.
 - Generated a web-based dashboard for intuitive data analysis and visualization.